

DATA ENGINEER SQL INTERVIEW SOLUTIONS

1. Combine Two Tables

TABLE: PERSON

personId	lastName	firstName
1	Wang	Allen
2	Alice	Bob

TABLE: ADDRESS

addressId	personId	city	state
1	2	New York City	New York
2	3	Leetcode	California

QUESTION

Write a SQL query to report the first name, last name, city, and state of each person in the Person table.

Rules:

- If the address of a personId is not present in the Address table, report null instead.
- Return the result table in any order.

EXPECTED OUTPUT

firstName	lastName	city	state
Allen	Wang	Null	Null
Bob	Alice	New York City	New York

Reference Solution

```
SELECT
    p.firstName,
    p.lastName,
    a.city,
    a.state
FROM Person p
LEFT JOIN Address a
    ON p.personId = a.personId;
```

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2. Employees Earning More Than Their Managers

TABLE: EMPLOYEE

id	name	salary	managerId
1	Joe	70000	3
2	Henry	80000	4
3	Sam	60000	Null
4	Max	90000	Null

QUESTION

Write an SQL query to find the employees who earn more than their managers.

Rules:

- managerId is the id of the employee's manager.
- Return the result table in any order.

EXPECTED OUTPUT

Employee
Joe

Reference Solution

```
SELECT
    e.name AS Employee
FROM Employee e
JOIN Employee m
    ON e.managerId = m.id
WHERE e.salary > m.salary;
```

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3. Duplicate Emails

TABLE: PERSON

id	email
1	a@b.com
2	c@d.com
3	a@b.com

QUESTION

Write an SQL query to report all the duplicate emails.

Rules:

- Emails will only consist of lowercase letters.
- Return the result table in any order.

EXPECTED OUTPUT

Email
a@b.com

Reference Solution

```
SELECT email AS Email
FROM Person
GROUP BY email
HAVING COUNT(email) > 1;
```

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4. Customers Who Never Order

TABLE: CUSTOMERS

id	name
1	Joe
2	Henry
3	Sam
4	Max

TABLE: ORDERS

id	customerId
1	3
2	1

QUESTION

Write an SQL query to report all customers who never order anything.

Rules:

- Return the result table in any order.

EXPECTED OUTPUT

Customers
Henry
Max

Reference Solution

```
SELECT
    c.name AS Customers
FROM Customers c
LEFT JOIN Orders o
    ON c.id = o.customerId
WHERE o.id IS NULL;
```

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5. Big Countries

TABLE: WORLD

name	continent	area	population	gdp
Afghanistan	Asia	652230	25500100	20343000
Albania	Europe	28748	2831741	12960000
Algeria	Africa	2381741	37100000	188681000

QUESTION

A country is big if it has an area of at least 3 million km², or a population of at least 25 million. Write an SQL query to report the name, population, and area of the big countries.

Rules:

- Return the result table in any order.

EXPECTED OUTPUT

name	population	area
Afghanistan	25500100	652230
Algeria	37100000	2381741

Reference Solution

```
SELECT
    name,
    population,
    area
FROM World
WHERE area >= 3000000
    OR population >= 25000000;
```

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6. Department Highest Salary

TABLE: EMPLOYEE

id	name	salary	departmentId
1	Joe	70000	1
2	Jim	90000	1
3	Henry	80000	2
4	Sam	60000	2
5	Max	90000	1

TABLE: DEPARTMENT

id	name
1	IT
2	Sales

QUESTION

Write an SQL query to find employees who have the highest salary in each of the departments.

Rules:

- If multiple employees have the same highest salary, return all of them.
- Do not use LIMIT.

EXPECTED OUTPUT

Department	Employee	Salary
IT	Jim	90000
IT	Max	90000
Sales	Henry	80000

Reference Solution

```
SELECT
    d.name AS Department,
    e.name AS Employee,
    e.salary AS Salary
FROM Employee e
JOIN Department d
    ON e.departmentId = d.id
WHERE (e.departmentId, e.salary) IN (
    SELECT departmentId, MAX(salary)
    FROM Employee
    GROUP BY departmentId
);
```

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7. Second Highest Salary

TABLE: EMPLOYEE

id	salary
1	100
2	200
3	300

QUESTION

Write an SQL query to report the second highest salary from the Employee table.

Rules:

- If there is no second highest salary, the query should report null.

EXPECTED OUTPUT

SecondHighestSalary
200

Reference Solution

```
SELECT MAX(salary) AS SecondHighestSalary
FROM Employee
WHERE salary < (
    SELECT MAX(salary)
    FROM Employee
);
```

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8. Nth Highest Salary

TABLE: EMPLOYEE

id	salary
1	100
2	200
3	300

QUESTION

Write an SQL query to report the nth highest salary from the Employee table.

Rules:

- If there is no nth highest salary, the query should report null.
- N is passed as a parameter to the function/procedure.

EXPECTED OUTPUT

getNthHighestSalary(2)
200

Reference Solution

```
CREATE FUNCTION getNthHighestSalary(N INT) RETURNS INT
BEGIN
  DECLARE M INT;
  SET M = N - 1;
  RETURN (
    SELECT DISTINCT salary
    FROM Employee
    ORDER BY salary DESC
    LIMIT 1 OFFSET M
  );
END
```


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9. Rank Scores

TABLE: SCORES

id	score
1	3.5
2	3.65
3	4.0
4	3.85
5	4.0
6	3.65

QUESTION

Write an SQL query to rank the scores. The ranking should be calculated according to the following rules:

Rules:

- The scores should be ranked from the highest to the lowest.
- If there is a tie between two scores, both should have the same ranking.
- After a tie, the next ranking number should be the next consecutive integer value.

EXPECTED OUTPUT

score	rank
4.0	1
4.0	1
3.85	2
3.65	3
3.65	3
3.5	4

Reference Solution

```
SELECT
    score,
    DENSE_RANK() OVER (ORDER BY score DESC) AS 'rank'
FROM Scores;
```

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10. Classes With at Least 5 Students

TABLE: COURSES

student	class
A	Math
B	English
C	Math
D	Biology
E	Math
F	Computer
G	Math
H	Math
I	Math

QUESTION

Write an SQL query to report all the classes that have at least five students.

Rules:

- Return the result table in any order.

EXPECTED OUTPUT

class
Math

Reference Solution

```
SELECT class
FROM Courses
GROUP BY class
HAVING COUNT(student) >= 5;
```

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11. Customer Placing Largest Number of Orders

TABLE: ORDERS

order_number	customer_number
1	1
2	2
3	3
4	3

QUESTION

Write an SQL query to find the customer_number for the customer who has placed the largest number of orders.

Rules:

- It is guaranteed that exactly one customer will have placed more orders than any other customer.

EXPECTED OUTPUT

customer_number
3

Reference Solution

```
SELECT customer_number
FROM Orders
GROUP BY customer_number
ORDER BY COUNT(order_number) DESC
LIMIT 1;
```

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12. Department Top Three Salaries

TABLE: EMPLOYEE

id	name	salary	departmentId
1	Joe	85000	1
2	Henry	80000	2
3	Sam	60000	2
4	Max	90000	1
5	Janet	69000	1
6	Randy	85000	1
7	Will	70000	1

TABLE: DEPARTMENT

id	name
1	IT
2	Sales

QUESTION

A company's executives are interested in seeing who earns the most money in each of the company's departments. A high earner in a department is an employee who has a salary in the top three unique salaries for that department.

Rules:

- Write an SQL query to find the employees who are high earners in each of the departments.

EXPECTED OUTPUT

Department	Employee	Salary
IT	Max	90000
IT	Joe	85000
IT	Randy	85000
IT	Will	70000
Sales	Henry	80000
Sales	Sam	60000

Reference Solution

```
WITH RankedSalaries AS (  
    SELECT  
        d.name AS Department,  
        e.name AS Employee,  
        e.salary AS Salary,  
        DENSE_RANK() OVER(  
            PARTITION BY e.departmentId  
            ORDER BY e.salary DESC  
        ) as rnk  
    FROM Employee e  
    JOIN Department d ON e.departmentId = d.id  
)  
SELECT Department, Employee, Salary  
FROM RankedSalaries  
WHERE rnk <= 3;
```

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13. Game Play Analysis I

TABLE: ACTIVITY

player_id	device_id	event_date	games_played
1	2	2016-03-01	5
1	2	2016-05-02	6
2	3	2017-06-25	1
3	1	2016-03-02	0
3	4	2018-07-03	5

QUESTION

Write an SQL query to report the first login date for each player.

Rules:

- Return the result table in any order.

EXPECTED OUTPUT

player_id	first_login
1	2016-03-01
2	2017-06-25
3	2016-03-02

Reference Solution

```
SELECT
    player_id,
    MIN(event_date) AS first_login
FROM Activity
GROUP BY player_id;
```

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14. Game Play Analysis II

TABLE: ACTIVITY

player_id	device_id	event_date	games_played
1	2	2016-03-01	5
1	2	2016-05-02	6
2	3	2017-06-25	1
3	1	2016-03-02	0
3	4	2018-07-03	5

QUESTION

Write an SQL query to report the device that is first logged in for each player.

Rules:

- Return the result table in any order.

EXPECTED OUTPUT

player_id	device_id
1	2
2	3
3	1

Reference Solution

```
SELECT
    player_id,
    device_id
FROM Activity
WHERE (player_id, event_date) IN (
    SELECT player_id, MIN(event_date)
    FROM Activity
    GROUP BY player_id
);
```

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15. Product Sales Analysis I

TABLE: SALES

sale_id	product_id	year	quantity	price
1	100	2008	10	5000
2	100	2009	12	5000
7	200	2011	15	9000

TABLE: PRODUCT

product_id	product_name
100	Nokia
200	Apple
300	Samsung

QUESTION

Write an SQL query that reports the product_name, year, and price for each sale_id in the Sales table.

Rules:

- Return the resulting table in any order.

EXPECTED OUTPUT

product_name	year	price
Nokia	2008	5000
Nokia	2009	5000
Apple	2011	9000

Reference Solution

```
SELECT
    p.product_name,
    s.year,
    s.price
FROM Sales s
JOIN Product p
    ON s.product_id = p.product_id;
```

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16. Product Sales Analysis II

TABLE: SALES

sale_id	product_id	year	quantity	price
1	100	2008	10	5000
2	100	2009	12	5000
7	200	2011	15	9000

TABLE: PRODUCT

product_id	product_name
100	Nokia
200	Apple
300	Samsung

QUESTION

Write an SQL query that reports the total quantity sold for every product id.

Rules:

- Return the resulting table in any order.

EXPECTED OUTPUT

product_id	total_quantity
100	22
200	15

Reference Solution

```
SELECT
    product_id,
    SUM(quantity) AS total_quantity
FROM Sales
GROUP BY product_id;
```


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17. Sales Analysis I

TABLE: PRODUCT

product_id	product_name	unit_price
1	S8	1000
2	G4	800
3	iPhone	1400

TABLE: SALES

seller_id	product_id	buyer_id	insert_date	quantity	price
1	1	1	2019-01-21	2	2000
1	2	2	2019-02-17	1	800
2	2	3	2019-06-02	1	800
3	3	4	2019-05-13	2	2800

QUESTION

Write an SQL query that reports the best seller by total sales price.

Rules:

- If there is a tie, report them all.

EXPECTED OUTPUT

seller_id
1
3

Reference Solution

```
WITH SellerSales AS (  
    SELECT seller_id, SUM(price) as total_sales  
    FROM Sales  
    GROUP BY seller_id  
)  
SELECT seller_id  
FROM SellerSales  
WHERE total_sales = (  
    SELECT MAX(total_sales)  
    FROM SellerSales  
);
```

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18. Not Boring Movies

TABLE: CINEMA

id	movie	description	rating
1	War	great 3D	8.9
2	Science	fiction	8.5
3	irish	boring	6.2
4	Ice song	Fantasy	8.6
5	House card	Interesting	9.1

QUESTION

Write an SQL query to report the movies with an odd-numbered ID and a description that is not "boring".

Rules:

- Return the result table ordered by rating in descending order.

EXPECTED OUTPUT

id	movie	description	rating
5	House card	Interesting	9.1
1	War	great 3D	8.9

Reference Solution

```
SELECT id, movie, description, rating
FROM Cinema
WHERE id % 2 = 1
      AND description != 'boring'
ORDER BY rating DESC;
```

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19. Total Sales Amount by Year

TABLE: PRODUCT

product_id	product_name
1	LC Phone
2	LC T-Shirt
3	LC Keychain

TABLE: SALES

product_id	period_start	period_end	average_daily_sales
1	2019-01-25	2019-02-28	100
2	2018-12-01	2020-01-01	10
3	2019-12-01	2020-01-31	1

QUESTION

Write an SQL query to report the total sales amount of each item for each year, with corresponding product_name, report_year, and total_amount.

Rules:

- Return the result table ordered by product_id and report_year.
- Dates are within 2018 to 2020.

EXPECTED OUTPUT

product_id	product_name	report_year	total_amount
1	LC Phone	2019	3500
2	LC T-Shirt	2018	310
2	LC T-Shirt	2019	3650
2	LC T-Shirt	2020	10
3	LC Keychain	2019	31
3	LC Keychain	2020	31

Reference Solution

```
WITH RECURSIVE DateRange AS (  
    SELECT product_id, period_start AS current_date,  
           period_end, average_daily_sales  
    FROM Sales  
    UNION ALL  
    SELECT product_id, DATE_ADD(current_date, INTERVAL 1 DAY),  
           period_end, average_daily_sales  
    FROM DateRange  
    WHERE current_date < period_end  
)  
SELECT  
    d.product_id,  
    p.product_name,  
    YEAR(d.current_date) AS report_year,  
    SUM(d.average_daily_sales) AS total_amount  
FROM DateRange d  
JOIN Product p  
    ON d.product_id = p.product_id  
GROUP BY d.product_id, p.product_name, report_year  
ORDER BY d.product_id, report_year;
```

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20. Average Selling Price

TABLE: PRICES

product_id	start_date	end_date	price
1	2019-02-17	2019-02-28	5
1	2019-03-01	2019-03-22	20
2	2019-02-01	2019-02-20	15
2	2019-02-21	2019-03-31	30

TABLE: UNITSSOLD

product_id	purchase_date	units
1	2019-02-25	100
1	2019-03-01	15
2	2019-02-10	200
2	2019-03-22	30

QUESTION

Write an SQL query to find the average selling price for each product.

Rules:

- average_price should be rounded to 2 decimal places.
- Return the result table in any order.

EXPECTED OUTPUT

product_id	average_price
1	6.96
2	16.96

Reference Solution

```
SELECT
    p.product_id,
    ROUND(SUM(u.units * p.price) / SUM(u.units), 2) AS average_price
FROM Prices p
JOIN UnitsSold u
    ON p.product_id = u.product_id
WHERE u.purchase_date BETWEEN p.start_date AND p.end_date
GROUP BY p.product_id;
```